MILITARY SPECIFICATION
DELAY LINE MX-1381/APM-66

This specification has been approved by the Bureau of Aeronautics, Department of the Navy

1. SCOPE AND CLASSIFICATION

1.1 Scope - The equipment (part of Radar Test Set, AN/APM-66) covered by this specification shall be designed to provide a quick convenient check by means of a delay line signal, on the operation of Radar Set AN/APN-22 as installed in an aircraft which is otherwise ready for flight. It shall be suitable for use on land or aboard ships.

1.2 Classification - The equipment covered by this specification shall be of one type and consist of the following:

Major Unit	Type Designation	Size (in.)	Weight (lb)	Applicable Paragraph
Delay Line	MX-1381/APM-66	21 x 6 x 10	20	3.3.1

2. APPLICABLE SPECIFICATIONS AND OTHER PUBLICATIONS

2.1 General - The following specifications and publications, of the issue in effect on the date of invitation for bids, shall form a part of this specification to the extent specified herein.

2.2 Specifications:

Military Specifications -

MIL-T-945	Test Equipment, for Use With Electronic Equipment; General Specification
JAN-P-658	Packaging and Packing of Electrical and Electronic Equipment and Spare Parts (Electronic, Electrical and Electromechanical)

2.3 Other Publications -

MIL-STD-129	Military Standard Markings of Shipments

2.4 Availability of Specifications and Publications:

(1) When requesting specifications and publications refer to both title and number. All requests should be made via the cognizant Inspector of Naval Material or Bureau of Aeronautics Representative.

(2) Copies of Military, Air Force-Navy Aeronautical, and Bureau of Aeronautics specifications may be obtained upon application to the Commanding Officer, Naval Air Station, Johnsville, Pennsylvania, Attn: Technical Records Division.

2.5 Precedence - When the requirements of the contract, this specification, or applicable subsidiary specifications are in conflict, the following precedence shall apply:

(1) Contract - The contract shall have precedence over any specification.

(2) This Specification - This specification shall have precedence over all applicable subsidiary specifications. Any deviation from this specification, or from subsidiary specifications where applicable, shall be specifically approved in writing by the Procuring Agency.

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4) Referenced Specifications - Any referenced specification shall have precedence over all applicable subsidiary specifications referenced therein. All referenced specifications shall apply to the extent specified.

3. REQUIREMENTS

3.1 Parts and Materials - In the selection of parts and materials, fulfillment of major design objectives shall be the prime consideration. In so doing, the following factors shall govern:

(1) Parts and materials approved by specification MIL-T-945 shall be given first consideration.

(2) When the contractors can demonstrate that the use of parts or materials which are in accordance with the specifications listed in MIL-T-945, will not fulfill the design objective because of size, weight, performance or other reasons, materials and parts shall be used which most nearly meet or exceed the performance requirements of the respective standard specification.

(3) If a nonstandard part is used, the meeting of general material requirements such as noninflammability, fungus resistance, nontoxicity, etc., and, the meeting of service conditions as required by Specification MIL-T-945 must be of prime consideration.

(4) In the selection of parts, the above considerations shall apply if the assembly is to be built for detail part maintenance. In the event the entire assembly is built to be expended, the assembly shall be constructed so that the unit as a whole will pass material, electrical and environmental tests, without consideration to the quality of the detail parts. Any assembly built to be expended shall be first approved by the Naval Air Development Center.

3.1.1 Nonstandard Part Approval and Material Approval - All nonstandard parts or materials (parts or materials which do not comply with the requirements of specifications listed in MIL-T-945) of an electrical nature, or methods of manufacture, and techniques and procedures involving their use and application shall be submitted to the Naval Air Development Center for approval as required by Specification MIL-T-945.

3.1.2 Design Objectives - The objectives in the design of the equipment specified herein are: accuracy, reliability, simplicity of operation, and minimum size and weight. The design should fulfill the requirements of Specification MIL-T-945.

3.2 General Requirements -

3.2.1 General - This equipment shall fulfill all applicable design and performance requirements of Specification MIL-T-945 unless otherwise specified herein.

3.2.2 Size and Weight -

3.2.2.1 Size - The over-all size of the equipment shall not be greater than 21 inches long (including handle) by 6 inches wide by 10 inches high.

3.2.2.2 Total Weight - The total weight of the equipment, excluding cables and mounting bases, shall not be greater than 20 pounds.

3.2.3 Service Conditions - The equipment shall operate satisfactorily under any combination of service conditions as specified in MIL-T-945 with the following exceptions:

3.2.3.1 Altitude - The altitude requirement for equipment operating shall be 3.4 inches of mercury, approximating an altitude of 50,000 feet.

3.2.3.2 Temperature - The performance requirements of this specification shall apply throughout the temperature range of -40° to +55° C. Nonoperating temperature conditions shall be as specified in MIL-T-945.

3.2.4 Stability -

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3.2.4.1 Electrical Stability - The equipment shall operate, as specified herein, for at least 200 hours either continuously or intermittently without necessity for readjustment of any controls other than those located on the front panel.

3.2.4.2 Mechanical Stability - All controls and adjustments shall be designed so that their position of adjustment will not be altered when the equipment is subjected to the specified service conditions.

3.2.5 Total Operating Life - The equipment shall have a minimum total operating life of 2000 hours with reasonable servicing and replacement of parts. Parts which may require replacement within this interval shall be specified by the manufacturer.

3.2.6 Interconnecting Cabling - This equipment shall be designed to operate without external interconnecting cabling.

3.2.7 Cables and Connectors - The delay line cable used in this equipment shall be type RG-55/U. Type UG-88/U connectors shall be used to join the cable to the antenna horns within the unit.

3.2.8 Power Requirements - This equipment shall require no power other than radio frequency energy from Radar Set AN/APN-22.

3.2.9 Overload Protection - Overload protection for this equipment shall not be provided in the equipment.

3.2.10 Moisture and Fungus Resistant Treatment - The equipment shall be fungus proofed by selection of parts and materials as non-nutrient for fungus, or the parts and materials shall be treated prior to their use in the equipment so that over-all spraying of the equipment is not necessary.

3.2.11 Identification of Equipment -

3.2.11.1 Nameplates and Nomenclature - Nameplates and nomenclature shall be in accordance with Specification MIL-T-945.

3.2.11.2 Marking - The marking of this unit shall be in accordance with paragraph 3.38 of Specification MIL-T-945. The marking on the cable connectors shall be stamped in black ink with a protective coating applied over the stamping.

3.2.12 Interchangeability - The interchangeability requirements as specified in Specification MIL-T-945 shall be extended to require interchangeability of units and component parts of this equipment with the latest model of any earlier production equipments. Such interchangeability shall be measured against a model, manufacturing drawings, or other technical information as provided by the Naval Air Development Center for that purpose.

3.2.13 Provisions for Maintenance and Field Testing - Provisions for maintenance and field testing shall meet the requirements of Specification MIL-T-945.

3.2.14 Performance -

3.2.14.1 General - The equipment shall be designed so that it will meet the requirements of this specification and Specification MIL-T-945.

3.2.14.2 Operation - Delay Line MX-1381/APM-66 shall provide a delay line signal equivalent to a 165-foot altitude signal, when coupled to and utilizing transmitter energy from Receiver-Transmitter RT-160/APN-22 which is a unit of Radar Set AN/APN-22.

3.2.14.3 Frequency Characteristics - This equipment shall operate in the frequency range from 4200 to 4400 megacycles.

3.2.14.4 Range - This equipment shall provide an artificial altitude signal of 165 feet when used in conjunction with Radar Set AN/APN-22.

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3.2.14.5 Tolerance - Artificial altitude signals from this unit shall have an accuracy ±5 feet when used with Radar Set AN/APN-22.

3.2.14.6 Indication - No indication of response to altitude signals shall be provided in this equipment. Indication of response to signals from this equipment shall be observed on the Height Indicator ID-257/APN-22 of Radar Set AN/APN-22.

3.3 Detail Design Requirements -

3.3.1 Delay Line, MX-1381/APM-66 - This equipment shall be designed to meet the following requirements:

3.3.1.1 Function - The delay line, when mounted against the antenna face of Receiver-Transmitter RT-160/APN-22, a unit of Radar Set AN/APN-22, shall present a load approximating the load presented by free space. It shall couple a signal, delayed in time and simulating 165 feet of altitude, as indicated in the AN/APN-22.

3.3.1.2 Form Factor and Mounting - This unit shall be housed in a rectangular combination case 21 by 6 by 10 inches, including protective cover. This unit shall be provided with two nonmetallic handles. No mounting fasteners shall be required.

3.3.1.3 Contents of Unit - This unit shall consist of a base plate, two antenna horns with radomes, and a delay line.

3.3.1.3.1 Base Plate - The base plate shall be made of conductive material, machined to 0.001 flat. The base plate shall have two apertures to coincide with the two antenna horns of the Receiver-Transmitter RT-160/APN-22. Two polarizing pins shall be provided on the base plate to correctly position the delay line antennas to the RT-160/APN-22 unit.

3.3.1.3.2 Radomes - The radomes shall be flush-fitted into the apertures of the base plate. The voltage standing-wave ratio of the antennas shall be improved by the inclusion of the radome. The radomes shall be sealed uniformly to the base plate.

3.3.1.3.3 Antenna Horns - Two identical antenna horns shall be provided, polarized to match the polarization of the RT-160/APN-22, and fitted with antenna probes to mate with UG-88/U connectors. The voltage standing-wave ratio of the antenna assembly shall be 1.05 to 1 over the frequency range from 4200 to 4100 megacycles when measured from the connector with the radome in place.

3.3.1.3.4 Delay Line - The delay line shall be a length of RG-55/U cable or equivalent, sufficient to provide 170 ± 3 feet of signal delay at the operating frequencies. The delay line shall be terminated with two UG-88/U connectors. The attenuation of the delay line at 4300 megacycles and 25° C shall be 100 ± 10 db. The delay line shall be installed in the case in a semipermanent manner. The delay line, when installed and connected to the antenna horns, shall present no discontinuities less than 60 db when measured from either horn.

3.3.1.3.5 Handles - Two handles shall be provided and fastened to the case. These handles shall be nonmetallic and of sufficient size to allow use of heavy gloves.

3.3.1.3.6 Case - The case shall be metallic and shall be capable of immersion in water to within one inch of the top without damage or effect on the electrical characteristics of the unit. The case shall be provided with nonskid pads on three surfaces to prevent scuffing or slipping on carrier deck. The case shall be provided with latching mechanism to secure the cover.

3.3.1.3.7 Cover - A detachable, drip-proof cover shall be provided. The cover shall contain an electrical schematic of the unit, permanently attached to the inside.

3.3.1.3.8 Controls - No controls shall be provided in this unit

4. SAMPLING, INSPECTION, AND TEST PROCEDURE

4.1 Classification - The equipment specified herein shall be subject to preproduction and inspection tests as follows:

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(1) Preproduction Tests - Preproduction tests shall be made on a sample(s) representative of the production equipments, to determine that the production equipments will meet the requirements of this specification and the applicable subsidiary specifications.

(2) Inspection Tests - Inspection tests shall be made on production equipments submitted for acceptance under the contract to determine conformance with the requirements of this specification and the applicable subsidiary specifications and to determine that the equipments are equivalent in performance and construction to the approved preproduction sample. Any production prior to approval of the preproduction sample shall be at the contractor's risk.

4.2 Preproduction Tests - Preproduction tests shall be conducted on two preproduction models representative of the production equipment as follows:

4.2.1 Scope of Preproduction Tests - Preproduction tests shall be conducted at the Naval Air Development Center and shall be in accordance with paragraph 4.2.1.2 of Specification MIL-T-945. The Naval Air Development Center, however, reserves the right to make additional preproduction tests other than those outlined in Specification MIL-T-945 to insure compliance with this specification and the contract.

4.2.2 Presubmission Tests - Prior to the submission of any models for preproduction testing, the contractor shall furnish facilities for and shall conduct tests as required in paragraph 4.2.4 of Specification MIL-T-945. In addition, the data required in paragraph 4.2.3 of Specification MIL-T-945 shall be submitted to the Naval Air Development Center for review and approval prior to submission of any models.

4.2.3 Approval of Preproduction Models - Approval of preproduction model shall be by the Naval Air Development Center upon satisfactory completion of all tests. No additional models shall be delivered prior to the approval of the preproduction model. Prefabrication of any equipments prior to the approval of the preproduction model will be at the contractor's own risk. One approved preproduction model will be returned to the contractor for his use in the fabrication and testing of the models to be submitted for acceptance under the contract. The preproduction model shall not be considered as one of the models under the contract, however, it may be reworked by the contractor and submitted for acceptance as a production model.

4.3 Inspection Tests - The contractor shall furnish all models and shall be responsible for accomplishing the required tests. When inspection is conducted at the contractor's plant, all inspection and testing shall be under the supervision of the cognizant Naval Inspector. Contractors not having laboratory testing facilities satisfactory to the Government Inspector shall engage the services of a commercial testing laboratory acceptable to the Naval Air Development Center. The contractor shall furnish test reports, in duplicate, showing quantitative results for all tests required by this specification, and signed by an authorized representative of the contractor or laboratory, as applicable. Acceptance or approval of material during course of manufacture shall in no case be construed as a guaranty of the acceptance of the finished product.

4.3.1 Scope of Inspection Tests - Inspection tests shall meet all the requirements of MIL-T-945, and shall be of the following two types:

(1) Individual Tests - Individual tests shall be conducted on each model submitted for acceptance under the contract.

(2) Sample Tests - Sample tests shall be conducted on the quantity of equipments as approved by the Naval Air Development Center. Models submitted for sample tests shall be selected by the Government Inspector after the equipment has passed the individual tests. Sample tests shall be more extensive than the individual tests and may include any of the tests listed under preproduction tests which are deemed necessary by the Government Inspector to determine that production models continue to be equivalent in performance and construction of the approved preproduction model.

4.4 Procedures - The procedures for conducting presubmission testing of preproduction models and inspection testing of production models shall be determined by the contractor. These test procedures shall be subject to review and approval by the Naval Air Development Center prior to the initiation of any testing.

4.5 Acceptance of Production Models - Acceptance of all production models shall be by the cognizant Naval Inspector.

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5. PACKAGING AND PREP

5.1 General - The equipmen
with the requirements of Specificat
Specification MIL-STD-129. Suital
in the exterior shipping container

6. NOTES

6.1 Ordering Information -
and date of this specification.

Patent Notice - When Government
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